

MATH 180 Strategies of Problem Solving

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August 20, 2025

Outline

- 1 SoPS Lecture 2
 - Building conjectures
 - Expert play
 - Wrapping up

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Pinching Pennies

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- arrange pennies in three piles
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- use small piles (≤ 4 coins per pile)

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- try to find **patterns**

Pinching Pennies: Debrief

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Let's combine the data ...

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Question

What do the losing positions have in common?

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Submit your conjecture:



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What do the losing positions have in common?

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There seems to be some mysterious math behind the scenes...

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$$(b_r \dots b_3 b_2 b_1 b_0)_2 = b_0 + 2b_1 + 4b_2 + 8b_3 + \dots + 2^r b_r$$

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Examples

$$2 = 10_2 = 010_2 = 0010_2$$

$$7 = 111_2, \quad 4 = 100_2, \quad 13 = 1101_2$$

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Question 3

What binary number is the same as the natural number 2025?

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Some losing states for **Player 1**

$(3, 2, 1), \quad (4, 6, 2), \quad (3, 3, 0)$

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Any patterns?

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Then **Player 2** should make the sum zero again.

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Whatever **Player 1** does, he will leave the sum nonzero.

Then **Player 2** should make the sum zero again.

Eventually we run out of stones, and that must happen on a turn when the sum becomes zero.

Wreck the professor

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Start with the initial piles (5, 7, 9).

Fight me!

Break time!

Activity (5 minutes)

- get up, stretch get a cup of coffee
- chat up your neighbor, introduce yourself

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- play the game through and see if you were right

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Question 1

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Question 1

What did we learn today about invariants?

Question 2

Faced with a new game, what strategy might we try using to determine perfect play?

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You have 21 pennies. On each turn, a player removes 1, 2, or 3 coins. The player who takes the last coin wins. Is the start a winning position for **Player 1** or **Player 2**?

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Activity (10 mins)

Discuss the puzzle with your neighbors. What is the solution?

Final thoughts

Reflection

- What similarities and differences did you see between the "proof" we did today, vs. last class?
- Explain which player has a winning strategy in the previous puzzle and carefully describe what their strategy should be.

Submit your responses by **Tonight!**